
Problem Framing

Decision Support Simulator for Soybean and Corn Producers
(MG, PR, MS, GO)

Course: Decision Analytics · Version: 4 (Final) · Language: English (EN-US)

PROJECT Bayer FieldView - ADMD6

DOCUMENT Problem Framing

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TRANSLATED FROM	Brazilian Portuguese (PT-BR)

1. BUSINESS CONTEXT

The Bayer Climate FieldView platform is a digital agriculture solution that collects massive volumes of data from machinery, satellites, weather stations, and soil sensors. The platform already captures high-quality data; the open question (and the heart of this project) is how to convert that data into practical, auditable decisions for the producer in the field.

Our scope has been intentionally restricted to soybean and corn producers in Minas Gerais (MG), Paraná (PR), Mato Grosso do Sul (MS), and Goiás (GO), four states that concentrate a relevant share of national grain production and present diverse climate regimes.

2. PROBLEM DEFINITION

The central problem is not a lack of data, but rather the difficulty of transforming available data into practical decisions and recommendations. The **Decision Agriculture** model that FieldView promotes is currently stalled by a manual, decentralized analysis process. We have structured this problem around three pillars:

2.1 Data Fragmentation and Overload

Information reaches the producer in an isolated form, delivered through separate tables, charts, and reports for each indicator. Because the volume of variables is massive, the producer must open each item individually and filter what is relevant on their own. There is no consolidated, decision-oriented view that brings together the pertinent variables at the moment of choice.

2.2 The Interpretation Gap: The Missing "Why"

Even with high-quality data, producers lack tools that correlate different indicators and transform them into actionable insights. The TAPI explicitly acknowledges the difficulty of cross-referencing external sources (climate, satellite imagery) with internal sources (productivity maps, soil fertility) to explain *why* a given outcome occurred. Furthermore, results are typically presented in complex technical language (spreadsheets, agronomic reports), making interpretation difficult for producers without deep technical training.

2.3 Operational and Technological Inefficiency

In the absence of an integrated tool, many producers end up building their own correlations manually, using Excel, self-managed dashboards, or external consultancies that are often inconsistent. This process is impractical, error-prone, and far removed from the experience Bayer intends to deliver. The manual complexity creates operational risks and prevents producers from anticipating the real return on technical adjustments, such as machinery calibration and seed density, under conditions of climatic uncertainty.

CONSOLIDATED PROBLEM STATEMENT

The Decision Agriculture model is being held back by a manual, fragmented, and excessively technical analysis process. Soybean and corn producers in MG, PR, MS, and GO are unable to confidently anticipate the real impact of planting adjustments on productivity under uncertain climate scenarios, because the available data is neither correlated nor presented in a decision-ready format.

3. THE DECISION THE SIMULATOR WILL SUPPORT

Our solution is a crop simulator embedded in a dashboard that automates the correlation between agronomic and climatic variables, delivering structured and auditable recommendations. Rather than forcing the producer to read raw reports, the simulator answers a single practical question at the moment of decision:

"Given my state, soil type, crop, seed population, season, and planted area, what productivity (sc/ha [sacks per hectare]) can I realistically expect under a pessimistic, realistic, and optimistic climate scenario?"

3.1 User Journey: Three Screens

- **Screen 1: Home:** An onboarding screen that contextualizes the user on the platform: which states (MG, PR, MS, GO) and crops (soybean, corn) are covered, what the model does, and how to use it.
- **Screen 2: Inputs:** The producer fills in six metrics describing their planting scenario:
 - Location (state): MG, PR, MS, or GO.
 - Field soil type being simulated.
 - Crop: soybean or corn.
 - Seed population (total number of seeds the producer intends to plant).
 - Season [TN: *safrá* = primary growing season; *safrinha* = second-season crop, typically corn, planted after the primary soybean harvest]: the different climatic variations drive productivity significantly (dry vs. rainy, cold vs. warm).
 - Planted area in hectares: used to convert the model output into sc/ha.
- **Screen 3: Results:** The output is deliberately not a single number. The simulator returns three indicators, each across three climate scenarios:
 - **Plantability CV% (Coefficient of Variation, CV%):** A statistical indicator that measures the uniformity of seed spacing. A low CV% means high uniformity, fewer gaps and doubles, and therefore greater productivity.
 - **Plasticity:** The capacity of a genotype to adapt its morphology and physiology to environmental variation. The indicator shows how well the selected seed responds to the conditions of the chosen scenario.
 - **Actual Productivity (sc/ha):** The final yield, presented across three scenarios (pessimistic, realistic, optimistic) so that the producer can choose the one that best fits their risk profile.

3.2 Fixed Model Assumptions

Climate is treated as a fixed assumption drawn from the location: for each state, we use the historical climate record. When the producer selects a season, the simulator generates the three scenarios based on past occurrences of that season in that state. This keeps the recommendation auditable for each given state.

3.3 One-Line Project Definition

A decision support tool that enables soybean and corn producers in MG, PR, MS, and GO to simulate their planting variables and maximize productivity (sc/ha), accounting for climatic variation and uncertainty scenarios.

3.4 Mental Modeling and Communication Framings

The success of the tool depends on how the recommendation is "packaged" for the decision-maker's cognition:

Framing Type	Description
Attribute Framing	<i>"With our simulator, you will have a strategic north for your plantation, with simulated planting results and historical comparisons, without having to spend seeds and raw materials."</i> (Positive framing on precision gain.)
Goal Framing	<i>"Without the clear scenario view our simulator provides, your planting is a shot in the dark, subject to invisible losses."</i> (Framing on loss aversion and waste avoidance.)
Risk-Choice Framing (Decision Logic): Optimistic Profile (Maximax)	The simulator highlights the maximum productive potential of the harvest.
Risk-Choice Framing: Conservative Profile (Maximin)	The tool focuses on which adjustment guarantees survival and the smallest possible loss in the pessimistic scenario.
Risk-Choice Framing: Opportunity Profile (Minimax Regret)	The tool focuses on minimizing the "regret" of a wrong decision. The simulator identifies the choice with the smallest gap between the outcome obtained and the best possible outcome in any scenario. This profile targets the producer seeking balance: they do not want to "leave money on the table" if conditions are favorable, but also want to avoid the opportunity cost of having been too aggressive should conditions fail.

4. STAKEHOLDERS AND INFORMATION NEEDS

Stakeholder	Primary Need
Soybean/corn producer (primary user)	A fast, visual answer to "how much will I harvest if I plant this way under these conditions?" without having to read technical reports.
Agronomic consultants	Use the simulator to support client recommendations with auditable scenarios.
Bayer Product Team	Platform owners benefit from increased engagement and clear value delivery from existing data assets.

5. SUCCESS CRITERIA

Criterion	Target
Model accuracy	More than 80% agreement between simulated and historically observed productivity.
Stickiness <i>[TN: stickiness = the capacity of a product, service, or brand to retain customers and encourage repeated, habitual use over time]</i>	More than 15%.
Visual simplicity	70% reduction in visual complexity compared to standard FieldView reports.

6. GOING BEYOND ("IR ALÉM")

6.1 Cost of the Current Problem

Manual correlation in Excel consumes hours of consulting time and leads to suboptimal seed density choices. A conservative estimate places the avoidable loss at approximately R\$ 45,000 per mid-sized farm per cycle, accounting for productivity loss due to poorly calibrated planting decisions and consulting hours spent redoing analyses.

6.2 AgTech Benchmarking

Players such as John Deere Operations Center and Trimble Ag Software are already advancing toward prescriptive recommendations, but their outputs remain technical and machinery-centric. Our differentiator is delivering a producer-friendly recommendation, based on scenarios and expressed in sc/ha, with three explicit risk levels.

6.3 Risk Assessment

Risk Category	Description	Mitigation
Data Risk	Historical climate series may have gaps for specific micro-regions.	Mitigated by state-level aggregation.
Model Risk	Overfitting to past climates.	Mitigated by cross-validation (holdout) and the three-scenario output, which explicitly communicates uncertainty.
Adoption Risk	Producers may distrust a "black box."	Mitigated by auditable assumptions (every scenario traces back to historical data).
Operational Risk	Integration with FieldView data pipelines.	Mitigated by starting with a standalone simulator that consumes exported data.

7. DEFINITIVE PROJECT SUMMARY

How can a soybean or corn producer in MG, PR, MS, or GO anticipate the actual productivity (sc/ha) of their planting decisions, through pessimistic, realistic, and optimistic climate scenarios, without relying on manual spreadsheets or highly technical reports?

8. GENERAL CONCLUSION

This Problem Framing artifact consolidates the transition from a data-centric view to a business-decision-oriented view within Bayer FieldView. By transforming fragmented indicators into a scenario simulator for MG, PR, MS, and GO, we create a cognitive bridge that enables producers to manage climatic uncertainty with statistical support rather than intuition. With scope focused on profitability per hectare and well-defined success criteria, the project is ready for the modeling phase, ensuring that the technology is converted into an essential, auditable tool for planting strategy across the Brazilian field.